

Problem 16.3.15

Evaluate counterclockwise.

$$\oint_C \tan 2\pi z \, dz, \quad C : |z - 0.2| = 0.2$$

Solution

The function $f(z) = \tan 2\pi z = \frac{\sin 2\pi z}{\cos 2\pi z}$ has simple poles at

$$\cos 2\pi z = 0 \implies 2\pi z = \frac{\pi}{2} + n\pi \implies z = \frac{1}{4} + \frac{n}{2}, \quad n \in \mathbb{Z}$$

Only $z_0 = \frac{1}{4}$ lies in C ,

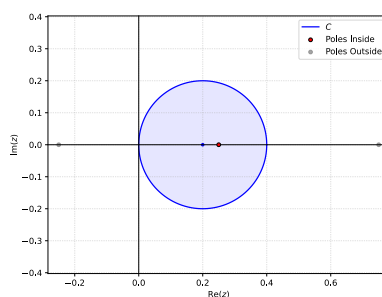


Figure 1: Poles

$$\operatorname{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = \frac{\sin 2\pi z}{-2\pi \sin 2\pi z} = -\frac{1}{2\pi}$$

The integral is then in the form,

$$\oint_C f(z) \, dz = 2\pi i \operatorname{Res}_{z=z_0} f(z)$$

$$\oint_C \tan 2\pi z \, dz = 2\pi i \operatorname{Res}_{z=\frac{1}{4}} f(z) = \boxed{-i}$$

Problem 16.3.17

Evaluate counterclockwise.

$$\oint_C \frac{e^z}{\cos z} dz, \quad C : \left| z - \frac{\pi i}{2} \right| = 4.5$$

Solution

The function $f(z) = \frac{e^z}{\cos z}$ has simple poles at

$$\cos z = 0 \implies z = \frac{\pi}{2} + n\pi, \quad n \in \mathbb{Z}$$

Poles at $z_{1,2} = \pm \frac{\pi}{2}$ lie within the region,

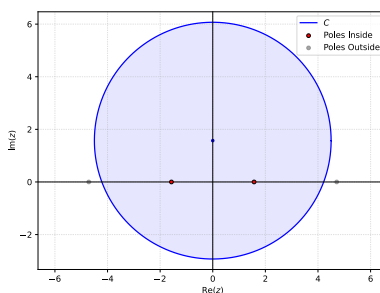


Figure 2: Poles

$$\operatorname{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = -\frac{e^{z_0}}{\sin z_0}$$

$$\operatorname{Res}_{z=\frac{\pi}{2}} f(z) = -\frac{e^{\frac{\pi}{2}}}{\sin \frac{\pi}{2}} = -e^{\frac{\pi}{2}}$$

$$\operatorname{Res}_{z=-\frac{\pi}{2}} f(z) = -\frac{e^{-\frac{\pi}{2}}}{\sin -\frac{\pi}{2}} = e^{-\frac{\pi}{2}}$$

The integral is then in the form,

$$\oint_C f(z) dz = 2\pi i \sum_{j=1}^k \operatorname{Res}_{z=z_j} f(z)$$

$$\oint_C \frac{e^z}{\cos z} dz = 2\pi i \left(-e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}} \right) = \boxed{-4\pi i \sinh\left(\frac{\pi}{2}\right)}$$

Problem 16.3.19

Evaluate counterclockwise.

$$\oint_C \frac{\sinh z}{2z - i} dz, \quad C : |z - 2i| = 2$$

Solution

The function $f(z) = \frac{\sinh z}{2z - i}$ has simple poles at

$$2z - i = 0 \implies z = \frac{i}{2}$$

The pole at $z_0 = \frac{i}{2}$ lies within the region,

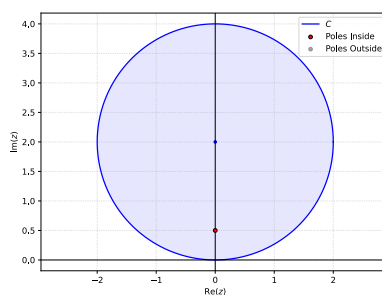


Figure 3: Poles

$$\text{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = \frac{1}{2} \sinh z_0$$

$$\text{Res}_{z=\frac{i}{2}} f(z) = \frac{1}{2} \sinh \frac{i}{2} = \frac{i}{2} \sin \frac{1}{2}$$

The integral is then in the form,

$$\oint_C f(z) dz = 2\pi i \text{Res}_{z=z_0} f(z)$$

$$\oint_C \frac{\sinh z}{2z - i} dz = 2\pi i \left(\frac{i}{2} \sin \frac{1}{2} \right) = \boxed{-\pi \sin \frac{i}{2}}$$

Problem 16.3.21

Evaluate counterclockwise.

$$\oint_C \frac{\cos \pi z}{z^5} dz, \quad C : |z| = \frac{1}{2}$$

Solution

The function $f(z) = \frac{\cos \pi z}{z^5}$ has a 5th-order pole at

$$z^5 = 0 \implies z = 0$$

The pole at $z_0 = 0$ lies within the region,

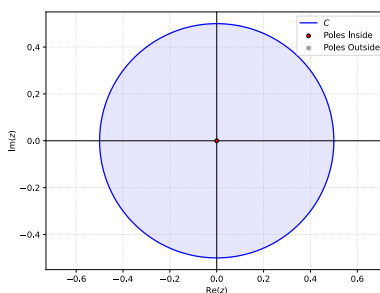


Figure 4: Poles

$$\operatorname{Res}_{z=0} f(z) = \frac{1}{4!} \lim_{z \rightarrow 0} \left\{ \frac{d^4}{dz^4} [z^4 f(z)] \right\} = \frac{1}{24} \frac{d^4}{dz^4} [\cos \pi z] \Big|_{z=0} = \frac{\pi^4}{24} \cos \pi z \Big|_{z=0} = \frac{\pi^4}{24}$$

The integral is then in the form,

$$\oint_C f(z) dz = 2\pi i \operatorname{Res}_{z=z_0} f(z)$$

$$\oint_C \frac{\cos \pi z}{z^5} dz = 2\pi i \left(\frac{\pi^4}{24} \right) = \boxed{\frac{i\pi^5}{12}}$$

Problem 16.4.1

Evaluate.

$$\int_0^\pi \frac{2}{k - \cos \theta} d\theta$$

Solution

Problem 16.4.7

Evaluate.

$$\int_0^{2\pi} \frac{a}{a - \sin \theta} d\theta$$

Solution

Problem 16.4.9

Evaluate.

$$\int_0^{2\pi} \frac{\cos \theta}{13 - 12 \cos 2\theta} d\theta$$

Solution

Problem 16.4.11

Evaluate.

$$\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2} dx$$

Solution

Problem 16.4.15

Evaluate.

$$\int_{-\infty}^{\infty} \frac{x^2}{x^6 + 1} dx$$

Solution

Problem 16.4.17

Evaluate.

$$\int_{-\infty}^{\infty} \frac{\sin 3x}{x^4 + 1} dx$$

Solution

Problem 16.4.19

Evaluate.

$$\int_{-\infty}^{\infty} \frac{1}{x^4 - 1} dx$$

Solution

Problem 16.4.21

Evaluate.

$$\int_{-\infty}^{\infty} \frac{\sin x}{(x-1)(x^2+4)} dx$$

Solution

Problem 16.4.25

Find the Cauchy principal value.

$$\int_{-\infty}^{\infty} \frac{x+5}{x^3-x} dx$$